

External Interface for heinrichs weikamp Dive-Computers with hwOS (OSTC3, OSTC cR and OSTC 2) and OSTC4/5

Last Update: 250412

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Thanks to: Pascal Pelmont, Artur Wróblewski, Sven Knoch, Jef Driesen, JD Gascuel, Dirk Hohndel, Lars Oestreicher, Linus Torvalds

hwOS Computers with USB:

The USB interface is equipped with an USB Chip (FT232) from FTDI. You can get Windows/Linux drivers from here: <http://www.ftdichip.com/Drivers/VCP.htm> (current Version is 2.08.24)

With Linux Kernel >2.6.9, these drivers are included in the Kernel.

In most cases, with Windows7 or Windows 8, these drivers will install automatically.

After installation a virtual COM port can be found in your windows hardware control panel.

You can change the COM port number there if required.

hwOS Computers with Bluetooth and OSTC4/5:

The uses Bluetooth chipset creates a virtual COM port, with Bluetooth 4.0 (LE) the device is compatible to "Stollman's Terminal IO Profile". See <http://www.stollmann.de/de/software/bluetooth-bluetooth-low-energy/terminal-io-profile.html>, newer OSTC use the u-connectXpress Bluetooth Low Energy Serial Port Service, see "UBX-16011192"

Software

The data rate is 115200 Baud, 8 Databits, 1 Stopbit, no parity, (115200, N, 8, 1)

16 and 24bit Values are transmitted in little Endian format (LOW:HIGH or (LOW:HIGH:UPPER)

Transfer protocol

Transfer is bi-directional and is initiated from the PC software by sending a command byte to the OSTC

- A command echo is send for every command
- Wait for answers/error bytes before starting new command!
- Keep COM port open until all transfers are done.
- After 120 seconds idle time the OSTC exits the COM mode – it's recommended to use a timeout lower then 120seconds and exit the COM mode from the PC application before the OSTC does (See Command 0xFF for details).
- hwOS:Right / OSTC4/5: Left .. button will abort COM mode when in idle loop.
- OSTC sends a 0x4D if it's ready for a new command.

Starting the communications with the OSTC:

- [Open COM Port]
- Enter Download mode: 0xBB
- Answer from OSTC:
- 0xBB (Command Echo)
- 0x4D -> OSTC will show "Download mode enabled"

Type * OSTC4/5 only † hwOs only	Command	Input	Output (w/o command echo and Ready/Done byte)	Ready/Done.
Start communication	0xBB	N	N	0x4D
Hardware + features	0x60	N	Data (6)	0x4D
Send headers	0x61	N	Data (65536)	0x4D

Send compact headers	0x6D	N	Data (4096)	0x4D
Set clock & date	0x62	Data (6)	N	0x4D
Custom text (1)	0x63	Data (60)	N	0x4D
Get dive profile	0x66	Data (1)	(Data n)	0x4D
Version/Identity (2)	0x69	N	Data (64)	0x4D
Firmware Details	0x6B	Data(1) = id	Data(4)	0x4D
Display text	0x6E	Data (16)	N	0x4D
Get hardware descriptor	0x6A	N	Data (1)	0x4D
Reset all settings	0x78	N	N	0x4D
Write setting (3)	0x77	See "Read/Write Settings" Data(6)	N	0x4D
Read setting (3)	0x72	Data(1) = id	See "Read/Write Settings" Data(4)	0x4D
Read limits and default	0x70	Data(1) = id	Data(4 or 7)	0x4D
Exit communication	0xFF	N	N	N

Note 1: OSTC4/5: automatic internal line break after every 12 characters, if no \n or \l is part of the string before. Only 4 x 12 = 48 characters are displayed

Note 2: OSTC4/5: limited version info, only major and sub-major version number, use 0x6B for complete info

Note 3: OSTC4/5: fixed length of 6 bytes for write and 5 bytes for read

command 0x60 („“): hardware descriptor + feature descriptor

Output OSTC: 0x60 (command echo)

Output OSTC: Hardware descriptor High byte (always 00 at the moment)

Output OSTC: Hardware descriptor Low byte (see table 0x6A)

Output OSTC: Feature descriptor High byte (always 00 at the moment)

Output OSTC: Feature descriptor Low byte (see table below)

Output OSTC: Model descriptor byte (see table below)

Output OSTC: Final 0x4D

Resulting feature bytes:

OSTC Model by hw descriptor	Feature Low byte (HEX)	Feature
OSTC 4	0x01	Bonex Scooter Version
standard / all	0x00	standard

Resulting model byte:

Specific OSTC Model	Feature Low byte (HEX)
<unknown>	0x00
OSTC 4	0x43

command 0x61 („a“): download all 256 dive headers from the OSTC

Output OSTC:

- 0x61 (Command Echo)

The complete TOC (Table of Contents) memory block will be send, no matter how many dives are stored. One TOC entry is the header of one dive stored in the OSTC's memory.

The Header does not contain the detailed profile for the dive.

The total of 256 headers (256 dives, max. in the OSTC Dive-Computer) will be send. No matter if all 256 dives are in

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the memory already used. Unused headers consists of 0xFF's only and must be ignored in the software.

Header size is 256byte/dive → A total of 65536Bytes will be transferred.

Download starts with dive #0 and ends with dive #255. Depending on how many dives made with the OSTC yet, this list is not sorted to the date and/or time of the dives. Sort by using date/time information from the header! Keep number (0-255) in your list, it's needed to get the dive profile for a particular dive. See command “f” for details.

Header details:

Byte Number	Length	Description	Data	Comment
0	2	Dive Header start	0xFA, 0xFA	For internal use
2	3	Internal pointer to the begin of the profile data	24bit Address	For internal use
5	3	Internal pointer to the end of the profile data	24bit Address	For internal use
8	1	Logbook-Profile version	8bit Version	Currently only 0x23 or 0x24
9	3	Length of profile data for this dive	24bit Length	Useful for parsing profile
12	3	Date of Dive at end (or start!) (YYMMDD)	3Byte Date	Year is 20xx, See Note 11
15	2	Time at end (or start!) of Dive (HHMM)	2Byte Time	See Note 11
17	2	Max. Depth in mbar (Includes salinity offset!)	16bit Depth	See Note 1
19	3	Divetime (MMMMSS)	24bit Divetime	2Bytes Min, 1Byte Sec.
22	2	Min. Water temperature in 0.1°C	16bit Temp	Signed 16Bit
24	2	Surface air pressure before dive in mbar	16bit Pressure	30min before dive
26	2	Desaturation time after the dive in minutes	16bit Minutes	
28	4	Gas/Dil 1 (%O2, %He, Change Depth in m, Type)	4Byte Gas data	See Note 2,3
32	4	Gas/Dil 2 (%O2, %He, Change Depth in m, Type)	4Byte Gas data	See Note 2,3
36	4	Gas/Dil 3 (%O2, %He, Change Depth in m, Type)	4Byte Gas data	See Note 2,3
40	4	Gas/Dil 4 (%O2, %He, Change Depth in m, Type)	4Byte Gas data	See Note 2,3
44	4	Gas/Dil 5 (%O2, %He, Change Depth in m, Type)	4Byte Gas data	See Note 2,3
48	2	Firmware Version xx.yy	2Byte Version	
50	2	Battery voltage in mV after the dive	16bit Voltage	
52	1	Sampling rate in Seconds	1Byte	Also stored in profile section
53	2	CNS at beginning of dive in %	16bit CNS	
55	1	GF at beginning of dive in %	1Byte GF	
56	1	GF at end of dive in %	1Byte GF	
57	2	Configured logbook offset in the OSTC	16Bit Offset	
59	1	Unused	1Byte	Ignore
60	2	Setpoint 1 (Setpoint in cbar, Change depth in m)	2Byte SP Data	1 cbar = 0.01bar
62	2	Setpoint 2 (Setpoint in cbar, Change depth in m)	2Byte SP Data	
64	2	Setpoint 3 (Setpoint in cbar, Change depth in m)	2Byte SP Data	
66	2	Setpoint 4 (Setpoint in cbar, Change depth in m)	2Byte SP Data	
68	2	Setpoint 5 (Setpoint in cbar, Change depth in m)	2Byte SP Data	
70	1	Salinity Setting	1Byte Salinity	See Note 5

Byte Number	Length	Description	Data	Comment
71	2	Max. CNS in %	16bit CNS	
73	2	Average depth in mbar	16bit Depth	See Note 1
75	2	Total dive time in seconds (incl. shallow parts)	16bit Seconds	See Note 6
77	1	If GF deco model was used : GF_lo [%] else: Saturation multiplier [%]	1Byte Deco model info	
78	1	If GF deco model was used : GF_hi [%] else: Desaturation multiplier [%]	1Byte Deco model info	
79	1	Deco model used for this dive	1Byte Model	0= ZH-L16, 1= ZH-L16-GF, 2=VPM
80	2	Dive number (Internal total dive counter)	16bit Number	See Note 7
82	1	Dive mode used for this dive	1Byte Mode	See Note 8
83	16	N2 compartment desaturation time in min.	16Bytes	See Note 9
99	64	N2 compartments at end of dive in mbar	64Bytes	16*4Byte (Float), See Note 9
163	16	He compartment desaturation time in min.	16Bytes	See Note 9
179	64	He compartments at end of dive in mbar	64Bytes	16*4Byte (Float), See Note 9
243	1	Last decompression stop in m	1Byte Depth	User-Set value
244	1	Assumed distance to shown stop in 10cm	1Byte	Safety-margin parameter
245	2	hwHUD battery value at end of dive in mV	16bit Voltage	Zero when no HUD present
247	1	hwHUD last status	1Byte Status	See Note 10
248	6	Battery gauge registers at end of dive in $1 \cdot 10^{-9} \text{As}$	48bit Battery	Only for 3,6V Battery
254	2	Dive Header end	0xFB, 0xFB	

Note 1: Depth in meters = Depth in mbar* $g_0/1000$ ($g_0=9,80665\text{m/s}^2$)

Note 2: "First Gas" always has change depth = 0m, [List contains OC Gases for OC and Diluents for CC](#)

Note 3: Gas Types: 0=Disabled, 1=First, 2= Travel, 3= Deco, Diluent Type: 0=Disabled, 1=First, 2=Normal

Note 4: bit0-3: Divisor, 0=never, 1=every Sample, 2=every 2nd Sample, ...bit4-7: Information size (0-15)

Note 5: 0-5 [%]

Note 6: very shallow (<1m) parts of the dive are not counted in the divetime (Byte 19-21)

Note 7: [This number is increased by one with every dive made and the number stored here is the number of total dives at the time this dive was made. It's not related to the TOC entry.](#)

Note 8: 0= OC, 1= CC, 2= Gauge, 3= Apnea

Note 9: Compartment assignment according to *Bühlmann, Albert: Tauchmedizin; 4. Auflage [2002]*, values after dive

Note 10: bit0=1:HUD connection ok, bit1=1:HUD is calibrated , bit2=1:HUD Battery is low (<3000mV), bit3=1:Sensor 1 active, bit4=1:Sensor 2 active, bit5=1:Sensor 3 active, bit6=1:HUD calibrated through external interface (DR5/DRx), bit7=1:HUD received (valid) data through RS232 input from DR5/DRx

Note 11: With profile version 0x24 or later, hwOS stores the START of the dive, not the END as before

[Note: OSTC sends a 0x4D after transferring all headers successfully.](#)

Profile Details

The profile for each dive has a variable length.

First, a small header with the information divisors is send.

Small Header

Byte Number	Length	Description	Data	Comment
0	3	Length of profile data for this dive	24bit Length	Useful for parsing profile
3	1	Sampling rate in Seconds	1Byte	Also stored in header section
4	1	Number of Divisors in this dive ($n=7$)	1Byte	Each divisor has 3 Bytes, See Note 1
5 ($1*3+2$)	3	Sampling Divisor Temperature	3Byte	See Note 2
8 ($2*3+2$)	3	Sampling Divisor Deco/NDL Status	3Byte	See Note 2
11 ($3*3+2$)	3	Sampling Divisor Gradient Factor	3Byte	See Note 2
14 ($4*3+2$)	3	Sampling Divisor ppO2 Sensors	3Byte	See Note 2
17 ($5*3+2$)	3	Sampling Divisor Decoplan	3Byte	See Note 2
20 ($6*3+2$)	3	Sampling Divisor CNS profile	3Byte	See Note 2
23 ($n*3+2$)	3	Sampling Divisor Tank data	3Byte	See Note 2
($n+1*3+2$)		Future Profile versions may have more divisors		

Note 1: Inspect each divisor (0=no Divisors, 1=One Divisor, etc.)

Note 2: Each divisor: Byte0= Type, Byte1=Length, Byte2=Divisor

Type=0: Temperature

Type=1: Deco/NDL Status

Type=2: Gradient Factor

Type=3: ppO2 Sensors

Type=4: Decoplan

Type=5: CNS

Type=6: Tank Data

Now, the actual dive profile is send.

Byte ($n*3+5$) to Byte #“Length of profile data for this dive”:

Depth in mbar (LOW:HIGH, not including salinity offset), ProfileFlagByte, extended Information

...

2 Byte:

0xFD 0xFD (Profile and dive end)

ProfileFlagByte:

bit0-6: amount of additional bytes after the ProfileFlagByte (0-127)

bit7: EventByte Flag (At least one Event Byte is available)

EventByte:

bit0-3: Alarm type

0=No Alarm

1=SLOW

2=Deco Stop missed

3=unused

4=ppO Low Warning
 5=ppO High Warning
 6=Manual Marker
 7=Low Battery

Bit4: Manual Gas Set („GasX”)

Bit5: Gas Change (Gas1-5 or Dil 1-5, called “Gas 6”-”Gas10” in the menu)

Bit6: SetPoint Change (cbar)

Bit7: Another Eventbyte is available

EventByte 2 (Only included if Bit7 of Eventbyte 1 = 1)

Bit0: Switched to bailout

Bit1-6: unused (yet)

Bit7: Another Eventbyte is available...

Events and Extended Information will be stored strictly in the following order (future extensions will append at the end of the list for compatibility reasons!):

Events:

- EventByte (s)
- ManualGasSet: new values for Gas 6 (2 bytes: byte1=O2%, byte2=He%)
- Gas Change (1 byte: current Gas#)
- SetPoint Change (1Byte: new SetPoint in cbar)
- Bailout: new values for bailout gas (2 bytes: byte1=O2%, byte2=He%)

Extended Information:

Type	Length	Data	Standard Divisor	Comment
Temperature	2	16bit short signed in 0.1°C	6	
Deco/NDL Status	2	Byte1:=0 if within NDL, if in deco mode: depth of 1st stop Byte2: NDL in minutes or length of first stop in minutes)	6	
GF Value	1	1Byte GF in %	12	
PpO2-Sensors	9	Per Sensor three bytes. Byte0: Sensor ppO2 (in 0.01bar) Byte1-2: 16bit Sensor Voltage in 0.1mV steps	2	Divisor=0 in OC dives
Deco Plan	15	Last 15 stops in minutes	12	Make sure to take care of header Byte243, which can be 3, 4, 5 or 6m
CNS value	2	16bit CNS value in %	12	
Tank Data	0	Not used yet	0	

For calculation of the available information the following rules apply:

- First sample# is 1
- if divisor == 0 the information will never appear
- dataAvailable = (sample# modulo divisor) == 0

Sampling rate can vary between dives! Always use sample rate from Dive Header.

Profile memory is a „first-in first-out” memory, so older dives will be overwritten when there is no free space left.

Profile for dives, where no profile information is available, will be two bytes: 0xFD 0xFD

Examples:

Assume the following configuration:

Divisor Temperature=3

Divisor Deco/NDL Status=1

Divisor Gradient Factor=0

Divisor ppO2 Sensors=0

Divisor Decoplan=0

Divisor CNS=0

Divisor Tank Data=0

Sample 1:

0x64 0x00 0x02 0x00 0xA0

Depth = 1m

ProfileFlagByte = 2 bytes will follow, No Event

Deco/NDL = NDL 160 minutes

Sample 2:

0xC8 0x00 0x02 0x00 0xA0

Depth = 2m

ProfileFlagByte = 2 bytes will follow, No Event

Deco/NDL = NDL 160 minutes

Sample 3:

0x2C 0x01 0x04 0x32 0x00 0x00 0xA0

Depth = 3m

ProfileFlagByte = 4 bytes will follow, No Event

Temperature = 5°C

Deco/NDL = NDL 160 minutes

Sample 4:

0x5E 0x01 0x84 0x21 0x02 0x00 0x9f

Depth = 3.5m

ProfileFlagByte = 4 bytes will follow, Event available (bit7=1)

EventByte = Alarm SLOW, Stored Gas Change

GasChange = Gas#2

Deco/NDL = NDL 159 minutes

...

Other Commands

command 0x62 („b“): Sync clock and date with PC

Output OSTC: 0x62 (command echo)

Now, 6 bytes must be send to the OSTC

Byte1: Hours

Byte2: Minutes

Byte3: Seconds

Byte4: Month

Byte5: Day

Byte6: Year (20xx)

All bytes will be checked for plausibility and the clock is set at once. After a timeout of about 30ms, the transmission will be aborted

Note: OSTC sends a 0x4D after setting the clock successfully

command 0x63 („c“): Send the custom text (up to 60 bytes)

Output OSTC: 0x63 (command echo)

Now, up to 60 bytes must be send to the OSTC

Byte 1-60: Custom text string shown when in surface mode (NULL terminated, last byte =0x00)

After a timeout of about 30ms, the transmission will be aborted and the text saved.

Note: OSTC sends a 0x4D after the text is saved.

command 0x66 („f“): Download one dive (incl. Profile data)

Output OSTC: 0x66 (command echo)

Now, 1 byte must be send to the OSTC

Byte1: Number of dive to download (0-255)

This number is the internal TOC number in which the OSTC has stored this dive.

Now, the OSTC sends the header (256bytes) + the profile (Ending with 0xFD 0xFD).

Note: Header bytes 8-10 contain the profile length in bytes including final 0xFD 0xFD

Depending on the dive length and profile details you can expect about 3Bytes/second divetime (e.g. 21,6kByte for a one-hour dive).

command 0x69 („i“): Download OSTC identity

Output OSTC: 0x69 (command echo)

Output OSTC: serial number (low:high)

Output OSTC: firmware version (x.y)

Output OSTC: Custom text (60 chars) Note: Custom Text is NULL terminated, but here all 60bytes are send.

Output OSTC: Final 0x4D

command 0x6A („j“): Download OSTC hardware descriptor byte

Output OSTC: 0x6A (command echo)

Output OSTC: Hardware descriptor byte

Output OSTC: 0x4D

Bit 0	OSTC has rechargeable battery with battery management chip
Bit 1	OSTC has an ambient sensor
Bit 2	OSTC has analog inputs and S8 digital
Bit 3	OSTC has an digital optical input
Bit 4	OSTC has a BLE module
Bit 5	OSTC has 32bit dual core
Bit 6	Unused (=0)
Bit 7	Unused (=0)

Resulting hardware byte:

OSTC Model	Resulting hardware bytes (HEX)
OSTC 3	0x0A
OSTC 3+	0x1A
OSTC cR	0x05
OSTC Sport	0x12
OSTC 2	0x11
OSTC 4	0x3B

Note: OSTC with old firmware (without the 0x6A command) will respond a 0x4D only.

command 0x6B („k“): Download OSTC firmware details (OSTC 4 only)

Output OSTC: 0x6B (command echo)

Now, 1 bytes must be send to the OSTC, 0xFF for main firmware, 0xFE for RTE, 0x10 for FontPackage

Output OSTC: firmware version (x.y.z 'betaFlag')

Output OSTC: Final 0x4D

command 0x6E („n“): Display up to 16 characters in the COM mode

Output OSTC: 0x6E (command echo)

Now, up to 16 bytes must be send to the OSTC

Byte 1-16: String to show in COM mode

Note: This is useful for debugging and showing some status (e.g.: “Reading Dives”) before actually downloading data (Which may take up to 2minutes for a full memory download).

Text can be updated as often as required whenever the COM mode is in the idle state waiting for commands. As all other commands, this will also reset the timeout.

Note: OSTC sends a 0x4D after displaying the new text

command 0x6D („m“): Download all 256 “Compact headers” from the OSTC

Output OSTC: 0x6D (command echo)

The complete list of “Compact headers” will be send, no matter how many dives are stored.

The Compact header contains basic informations about the dive. With it's small amount of data the use of this command is recommended to determine which dive needs to be downloaded. Especially when downloading from Bluetooth LE enabled OSTC's.

The total of 256 compact headers (256 dives, max. in the OSTC Dive-Computer) will be send. No matter if all 256 dives are in the memory already used. Unused headers consists of 0xFF's only and must be ignored in the software. Header size is 16byte/dive → A total of 4096Bytes will be transferred.

Download starts with dive #0 and ends with dive #255. Depending on how many dives made with the OSTC yet, this list is not sorted to the date and/or time of the dives. Sort by using date/time information from the compact header! Keep number (0-255) in your list, it's needed to get the dive profile for a particular dive. See command “f” for details.

Compact header details:

Byte Number	Length	Description	Data	Comment
0	3	Length of profile data for this dive	24bit Length	Useful for parsing profile
3	3	Date of Dive at end (or start!) (YYMMDD)	3Byte Date	Year is 20xx, See Note 3
6	2	Time at end (or start!) of Dive (HHMM)	2Byte Time	See Note 3
8	2	Max. Depth in mbar (Includes salinity offset!)	16bit Depth	See Note 1

Byte Number	Length	Description	Data	Comment
10	3	Divetime (MMMMSS)	24bit Divetime	2Bytes Min, 1Byte Sec.
13	2	Total dive number	16bit Number	See Note 2
15	1	Logbook-Profile version	8bit Version	Only in 1.92 or later (Older firmwares send 0xFF instead)

Note 1: Depth in meters = Depth in mbar* $g_0/1000$ ($g_0=9,80665\text{m/s}^2$)

Note 2: This number is increased by one with every dive made and the number stored here is the number of total dives at the time this dive was made. It's not related to the TOC entry.

Note 3: With profile version 0x24 or later, hwOS stores the START of the dive, not the END as before

Note: OSTC sends a 0x4D after transferring all compact headers successfully.

command 0x70: Read type, min, default and max for specific setting from OSTC4/5

Output OSTC4/5:

Byte 0: Type unknown, int15, int8, deci,centi, mili, percent, sec, color, bool, enum, signed, sdeci, ssec, sint

for all but int15:

Byte 1: min

Byte 2: default

Byte 3: max

int15:

Byte 1: min LOW

Byte 2: min HIGH

Byte 3+4: default LOW, HIGH

Byte 5+6: max LOW, HIGH

command 0xFF: Exit COM mode

Output OSTC: 0xFF (command echo)

The OSTC displays “Exited” for two seconds and powers-down the COM chip.

Command 0x77 and 0x72: Read/Write Settings:

1. Byte (Read/Write)

Write: 0x77 („w“)

+2...5 Bytes Data

OSTC will send 0x4D when all bytes have been received

Read: 0x72 („r“)

+1 Byte Data

OSTC will send 1...4 Bytes information + 0x4D (OSTC4/5 always 4 Bytes)

Output OSTC: 1 Byte (command echo)

2. Byte	3. Byte	4. Byte	5. Byte	6. Byte
Gas1: 0x10	%O2	%He	Type (Note 1)	Change Depth [m]
Gas2: 0x11	%O2	%He	Type (Note 1)	Change Depth [m]
Gas3: 0x12	%O2	%He	Type (Note 1)	Change Depth [m]
Gas4: 0x13	%O2	%He	Type (Note 1)	Change Depth [m]
Gas5: 0x14	%O2	%He	Type (Note 1)	Change Depth [m]
Dil1: 0x15	%O2	%He	Type (Note 2)	Change Depth [m]
Dil2: 0x16	%O2	%He	Type (Note 2)	Change Depth [m]
Dil3: 0x17	%O2	%He	Type (Note 2)	Change Depth [m]
Dil4: 0x18	%O2	%He	Type (Note 2)	Change Depth [m]
Dil5: 0x19	%O2	%He	Type (Note 2)	Change Depth [m]
SP1: 0x1A	Setpoint [cbar]	Change Depth [m]		
SP2: 0x1B	Setpoint [cbar]	Change Depth [m]		
SP3: 0x1C	Setpoint [cbar]	Change Depth [m]		
SP4: 0x1D	Setpoint [cbar]	Change Depth [m]		
SP5: 0x1E	Setpoint [cbar]	Change Depth [m]		
CCR mode: 0x1F	0=Fixed SP, 1=Sensor			
Dive mode: 0x20	0=OC, 1=CC, 2=Gauge, 3=Apnea			
Deco type: 0x21	0=ZH-L16, 1=ZH-L16-GF, 2=VPM			
ppO2 max: 0x22	ppO2 max [cbar]			
ppO2 min: 0x23	ppO2 min [cbar]			
Future TTS: 0x24	FTTS [min]			
GF_low: 0x25	GF_low [%]			
GF_high: 0x26	GF_high [%]			
aGF_low: 0x27	aGF_low [%]			
aGF_high: 0x28	aGF_high [%]			
aGF selectable: 0x29 VPM conservatism	0=aGF disabled, 1=aGF can be selected underwater VPM conservatism 0-5			
Saturation: 0x2A	Saturation [%]			
Desaturation: 0x2B	Desaturation [%]			
Last deco: 0x2C	Last Stop [m]			

Brightness: 0x2D	0=Eco, 1=Medium, 2=Full			
Units: 0x2E	0=m/°C, 1=ft/°F			
Sampling rate: 0x2F	0=2s, 1=10s			
Salinity: 0x30	0-4 [%]			
Dive mode colour: 0x31	0=Standard, 1=Red, 2=Green, 3=Blue			
Language: 0x32	0=EN, 1=DE, 2=FR, 3=IT, 4=ES			
Date format: 0x33	0=MMDDYY, 1=DDMMYY, 2=YYMMDD			
Compass gain: 0x34	(Note 3)			
Pressure sensor offset: 0x35	In mbar (Note 4)			
Show a safety stop: 0x36	0=No, 1=Yes			
Set calibration gas: 0x37	%O2			
Setpoint fallback: 0x38	0=No fallback, 1=Fallback			
Flip screen: 0x39	0=normal, 1=180° flipped			
Left button sensitivity: 0x3A (Note 5) All buttons sensitivity	20-100 [%] 70-110 [%]			
Right button sensitivity: 0x3B (Note 5) Button Balance	Bits:FxLLCCRR			
Bottom Gas consumption: 0x3C	5-50 [l/min]			
Decompression Gas consumption: 0x3D	5-50 [l/min]			
MOD Warning: 0x3E Travel Gas consumption 0x3E	0=No, 1=Yes 5-50 [l/min]			
Dynamic ascend rate: 0x3F	0= Fixed to 10m/min, 1=variable speed			
Graphical speed indicator: 0x40	0=No, 1=Yes			
Always show ppO2: 0x41	0=No, 1=Yes			
Temperature sensor offset: 0x42	In 0.1°C (Note 4)			
SafetyStop Length: 0x43 (Note 6)	60-240 [s] 60-255 [s]			
SafetyStop Start Depth: 0x44 (Note 7)	21-61[cbar] (2,1-6,1m) 30-60[cbar] (3-6 m)			
SafetyStop End Depth: 0x45	19-39 [cbar] (1,9-3,9m)			
SafetyStop Reset Depth: 0x46	81-151 [cbar] (8,1-15,1m)			
logbook offset 0x47	0..9000 low byte	0..9000 high byte		
Extra display 0x71	0=off, 1=BigFont			
Custom View Center 0x72	0..8 (..9 Bonex Version)			
CV Center Fallback 0x73	0..20 sec			

Custom View Corner 0x74	1..7			
CV Corner Fallback 0x75	0..20 sec			

Note 1: 0=Disabled, 1=First, 2=Travel, 3=Deco

Note 2: 0=Disabled, 1=First, 2=Normal

Note 3: 0=230LSB/Gauss, 1=330LSB/Gauss, 2=390LSB/Gauss, 3=440LSB/Gauss, 4=660LSB/Gauss, 5=820LSB/Gauss, 6=1090LSB/Gauss, 7=1370LSB/Gauss (See HMC5883L Datasheet, Page 13, for details). It's recommended not to adjust the compass gain externally.

Note 4: signed int8 (two's complement), values between +/-20 are valid

Note 5: Only for devices with piezo-electric buttons,

lower values mean higher sensitivity (hwOS), lower values mean lower sensitivity (OSTC4/5)

0x3B: balance values: LL = left button, CC = center button, RR = right button, F= Reset to Factory Defaults

values: 0x00 = -10, 0x01 = 0/standard, 0x10 = +10, 0x11 = +20

Note 6: rounded up to next Minute, > 240 = 5 Minutes, will set Show a safety stop to 1

Note 7: rounded up to next Meter, range 3m - 6m

Note: OSTC sends a 0x4D after read or write of one setting.

Note: Use command 0x6E („n“): Display up to 16 characters in the COM mode to display some feedback on the OSTC screen when updating settings.

Example 1: Set Gas 3 to TX18/40 (First Gas)

Command: 0x77 (Write)

Output OSTC: 0x77 (command echo)

Command: 0x12, d'17', d'40', 0x01, 0x00 (Gas3, O2=17, He=40, Type=1, Change Depth=0)

Output OSTC: 0x4D (Command done)

Example 2: Set Gas 5 to TX50/10 (Deco Gas, 21m)

Command: 0x77 (Write)

Output OSTC: 0x77 (command echo)

Command: 0x14, d'50', d'10', 0x03, d'21' (Gas5, O2=50, He=10, Type=3, Change Depth=21)

Output OSTC: 0x4D (Command done)

Example 3: Set OSTC to CCR Mode with Fixed Setpoints

Command: 0x77 (Write)

Output OSTC: 0x77 (command echo)

Command: 0x20, 0x01 (Dive mode=1)

Output OSTC: 0x4D (Command done)

Command: 0x77 (Write)

Output OSTC: 0x77 (command echo)

Command: 0x1F, 0x00 (CCR Mode=0)

Output OSTC: 0x4D (Command done)

As soon as command 1 is done (0x4D from OSTC) the next command can be started.

command 0x78 („x“): Reset all settings

Output OSTC: 0x78 (command echo)

Output OSTC: Final 0x4D